PRODUCTS OF K-INJECTIVE COMPLEXES ARE K-INJECTIVE AND REPRESENT THE DERIVED PRODUCT

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Editorial note. Counted span is the author's own proof environment, source lines 9925-9961, following the statement at 9916-9923.

Editorial note. No omissions: the product representation is argued for an arbitrary test complex K.

Allowed standard prerequisites (editorial).

- Hom complex computing morphisms in K. same chapter construction; the complexes C and C_t displayed in the proof (9925-9940). Middle cohomology is Hom_K(K, I) (resp. Hom_K(K, I_t)).
- K-injective recognition and D-level Hom. lemma-K-injective (same chapter, cited at ~9955). Converts vanishing of Hom_K(K, I) for acyclic K into the D-level product formula.

Source reference. Original lemma 070I: sources/stacks-derived/pinned-original-source.tex, source lines 9849–9849. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. Let T be a set and for each $t \in T$ let $I_t^\bullet$ be a K-injective complex. If $I^n = \prod_t I_t^n$ exists for all n, then $I^\bullet$ is a K-injective complex. Moreover, $I^\bullet$ represents the product of the objects $I_t^\bullet$ in $D(\mathcal{A})$.*

Proof. Let $K^\bullet$ be an complex. Observe that the complex

$$C : \prod\nolimits_b \mathrm{Hom}(K^{-b}, I^{b-1}) \to \prod\nolimits_b \mathrm{Hom}(K^{-b}, I^b) \to \prod\nolimits_b \mathrm{Hom}(K^{-b}, I^{b+1})$$

has cohomology $\mathrm{Hom}_{K(\mathcal{A})}(K^\bullet, I^\bullet)$ in the middle. Similarly, the complex

$$C_t : \prod\nolimits_b \mathrm{Hom}(K^{-b}, I_t^{b-1}) \to \prod\nolimits_b \mathrm{Hom}(K^{-b}, I_t^b) \to \prod\nolimits_b \mathrm{Hom}(K^{-b}, I_t^{b+1})$$

computes $\mathrm{Hom}_{K(\mathcal{A})}(K^\bullet, I_t^\bullet)$. Next, observe that we have

$$C = \prod\nolimits_{t \in T} C_t$$

as complexes of abelian groups by our choice of I. Taking products is an exact functor on the category of abelian groups. Hence if $K^\bullet$ is acyclic, then $\mathrm{Hom}_{K(\mathcal{A})}(K^\bullet, I_t^\bullet) = 0$, hence C_t is acyclic, hence C is acyclic, hence we get $\mathrm{Hom}_{K(\mathcal{A})}(K^\bullet, I^\bullet) = 0$. Thus we find that $I^\bullet$ is K-injective. Having said this, we can use Lemma 070I to conclude that

$$\mathrm{Hom}_{D(\mathcal{A})}(K^\bullet, I^\bullet) = \prod\nolimits_{t \in T} \mathrm{Hom}_{D(\mathcal{A})}(K^\bullet, I_t^\bullet)$$

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and indeed $I^\bullet$ represents the product in the derived category. $\square$